Traffic Violations & Weather SQL Analysis

MySQL Portfolio Project

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Project Overview

This project analyzes Washington, DC traffic violations and weather conditions using MySQL. The goal is to identify citation trends, agency patterns, peak violation periods, fine revenue drivers, and possible weather-related impacts on traffic enforcement — demonstrating an end-to-end SQL workflow from schema design through advanced analytics.

Datasets Used

Dataset	Description	Records
Traffic Violations	Cleaned DC traffic citation records with violation details, fine amounts, agencies, dates, and hours	338
Weather	Cleaned weather observations including temperature, precipitation, humidity, and conditions	208

SQL Skills Demonstrated

- Database and table creation in MySQL
- CSV import, data validation, and data cleaning checks
- Joins between violations and weather tables using converted date fields
- Aggregations for KPI reporting and trend analysis
- Window functions including RANK(), DENSE_RANK(), and running totals

Business Questions Answered

- Which violation types occur most frequently?
- Which agencies issue the highest number of citations?
- What hours show the highest violation activity?
- Which violation categories generate the most fine revenue?
- How do weather conditions connect with traffic violation patterns?

Repository Structure

File / Folder	Purpose

`data/`	Contains sample traffic violation and weather CSV files
`sql/01_create_tables.sql`	Creates the database and base tables
`sql/02_data_cleaning.sql`	Validates row counts, null values, duplicates, and date formats
`sql/03_violation_analysis.sql`	Analyzes violations by type, hour, agency, and fine amount
`sql/04_weather_impact_analysis.sql`	Joins violations with weather data and analyzes conditions
`sql/05_window_functions.sql`	Uses ranking and running totals for advanced SQL analysis
`visuals/`	Stores screenshots of query outputs and rankings

Key Takeaways

- Demonstrates end-to-end SQL workflow: schema design, import troubleshooting, validation, joins, aggregations, and window functions.
- Traffic violations can be analyzed by time, agency, violation type, and fine amount to support enforcement trend reporting.
- Weather data can be integrated with citation records to explore contextual patterns across dates and conditions.
- The GitHub repository provides recruiter-visible proof of MySQL, data cleaning, KPI analysis, and business reporting skills.

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